

18th International Nectar Conference

The Effect of Job Accessibility on Employment Outcomes: Evidence from the Canadian National Survey on Transport Poverty

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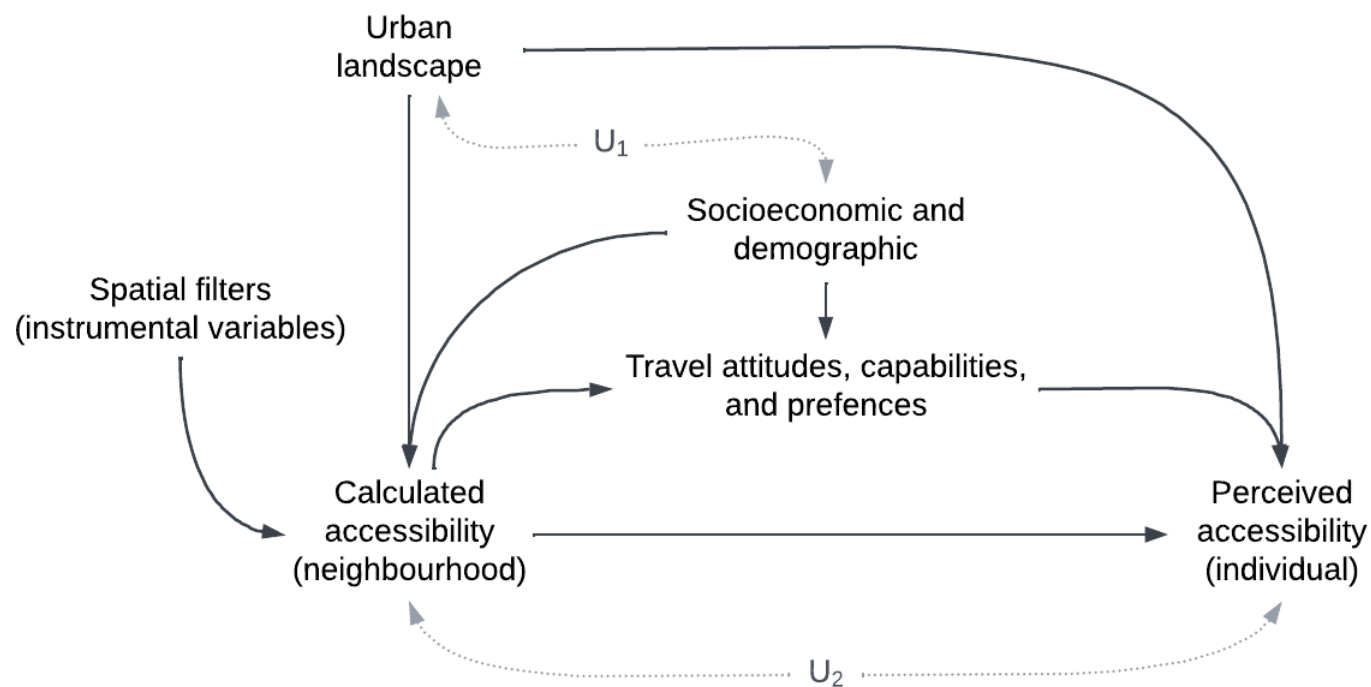
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Calculated vs perceived accessibility

- Often, calculated accessibility indicators deviate from perceived accessibility (Zhu and Lucas, 2025).
- Explanations for these differences are partially attributed to differences in individuals' travel attitudes, capabilities, and preferences (Pot, 2025; De Vos et al., 2025; van der Vlugt et al., 2022).
- The Mobilizing Justice Partnership (Tiznado-Aitken et al., 2025) recently released the Canadian National Survey on Transport Poverty (NSTP)

Objective: to evaluate the effect of calculated job accessibility on perceived employment reachability, using the microdata of the Canadian National Survey on Transport Poverty

Modelling approach



Directed Acyclic Graph (DAG) to model the effect of calculated accessibility on perceived accessibility.

Treatment:

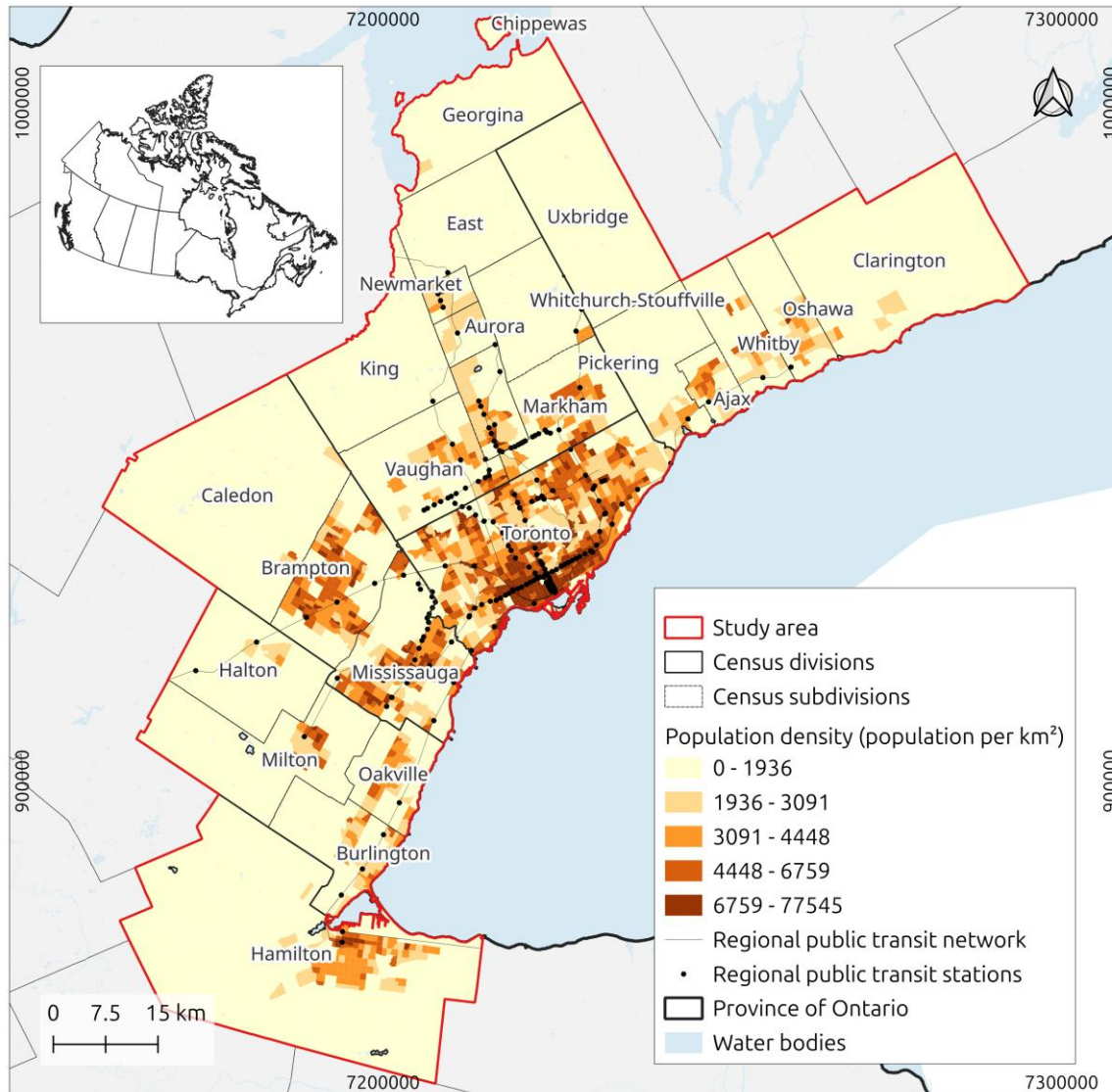
- Calculated accessibility

Controls:

- Gender, Disability, Indigenous or racialized, Temporary visa or refugee, Age
- Household size and income
- Number of cars, number of bikes, most used mode, match preferred and most used mode, public transport availability, monthly/annual transit pass
- Population density and population centre class
- *Travel time to reach work**

References for the control variables:

Delmelle et al., 2021; Jin & Paulsen, 2018; Hu, 2017; Johnson et al., 2017; Duarte et al., 2023; Bastiaanssen et al., 2021; Bastiaanssen et al., 2025.



Map of the study area.

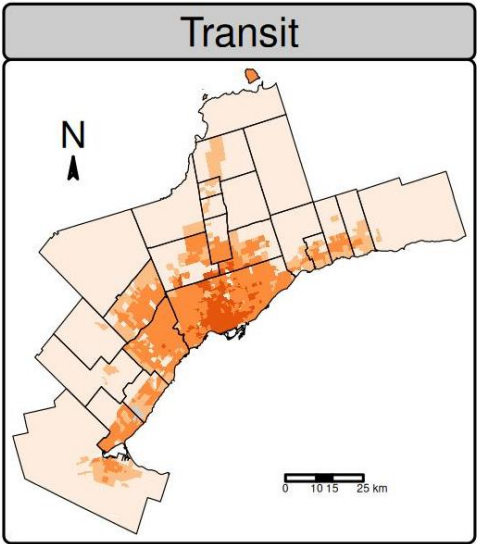
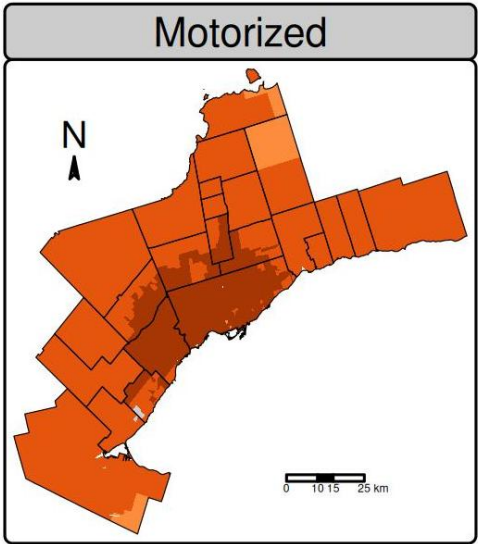
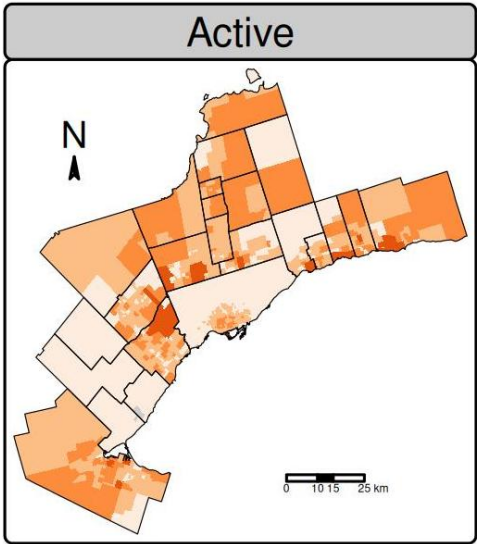
Greater Toronto and the Hamilton Area (GTHA)

25 Census Subdivisions within the Toronto Census Metropolitan Area and Hamilton Census Metropolitan Area
7,300 square kilometres, 7.2 million residents (Census, 2021)

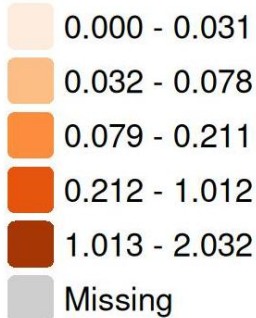
Accessibility models

- Input:
 - 3.8 million people in the labour force
 - 3 million job opportunities
 - Calibrated impedance functions (using the commute to work duration) to represent people's travel behaviour
 - Travel time with r5r (Pereira et al., 2021)
- *Multimodal spatial availability per person (Soukhov et al. 2024)*
 - Competitive and constrained
 - Multimodal: Active, Motorized, and Public transit
- Census tract as the spatial unit

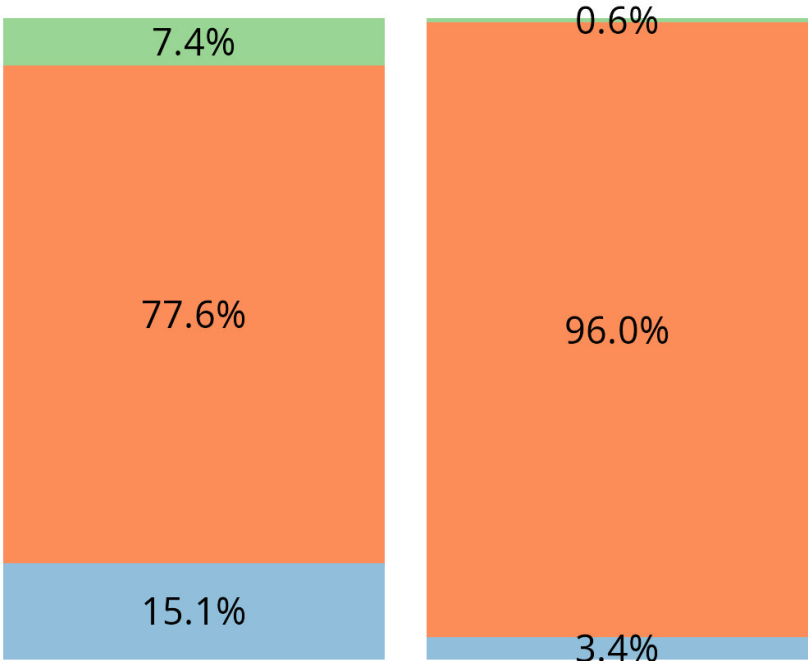
Multimodal Spatial Availability Per Person



Spatial availability per person



Mode	Median	Maximum
Active	0.04	2.03
Motorized	1.08	1.40
Public Transit	0.12	0.36



Population
by Mode
(Total: 3868120)

Spatial Availability
by Mode
(Total: 3043730)

Active Motorized Transit

Share of travel modes in the labour force and in the number of reached jobs.

Number of jobs reached per person by transportation mode for the study area
(Census Tract level, Jenks classification).

National Survey on Transport Poverty

To what extent do you agree with the following statements?

**0: Completely disagree
100: Completely agree**

Labour force N = 2,270; Weighted = 3,576,893	There are too few jobs available for me in my ideal commuting time	58.84 (27.71) ¹
	I have had to decline employment opportunities because of my transport situation	41.00 (31.62) ¹
Employed population N = 1,091; Weighted = 1,787,190	It takes too long to travel to work	54.04 (29.26) ¹
	With the transport options available to me, I can easily reach work	72.64 (22.58) ¹

Questions about perceptions of employment accessibility used in this study.

¹ Weighted Mean (SD)

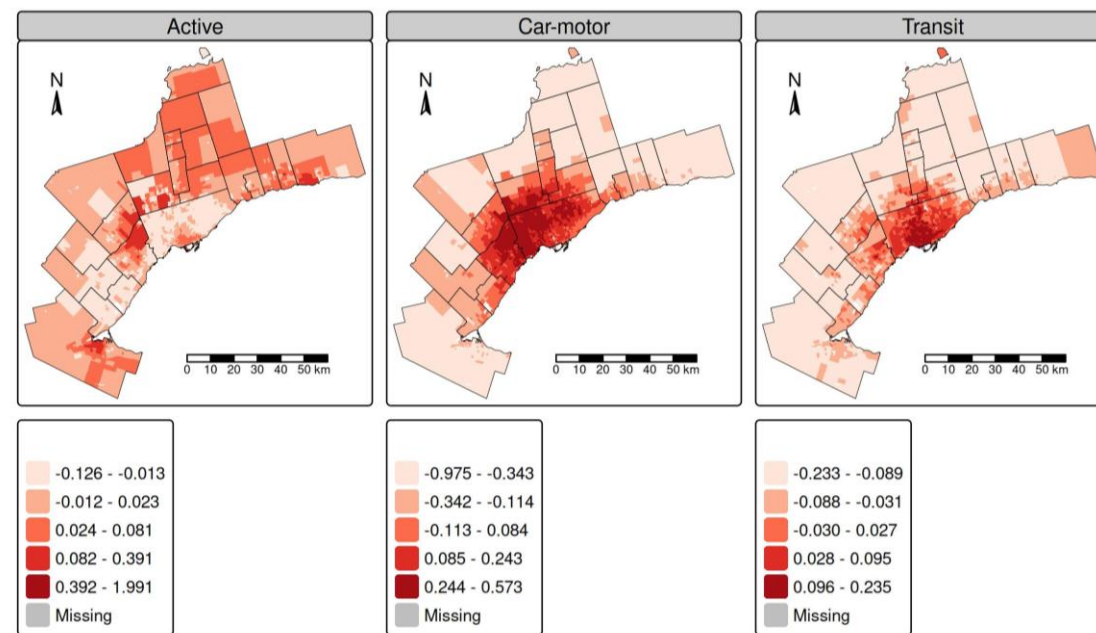
Modelling approach

<i>Perception</i>	<i>Most positive</i>	<i>Positive</i>	<i>Neutral</i>	<i>Negative</i>	<i>Most negative</i>
There are too few jobs available for me in my ideal commuting time	35	52	69	87	100
I have had to decline employment opportunities because of my transport situation	8	26	51	72	100
It takes too long to travel to work	21	51	65	82	100
With the transport options available to me, I can easily reach work	100	97	51	68	52

Weighted quintile thresholds used to categorize outcomes.

Modelling approach

- **Weighted linear** and **weighted ordinal categorical** regressions (Survey R Package of Lumley, 2011)
- Causality for accessibility:
 - Instrumental variables: spatial filters based (Le Gallo and Páez, 2013) on the calculated accessibility
 - Instruments are strongly correlated to the calculated accessibility and weakly correlated to the outcomes
 - Linear models: Two-Stage Residual Inclusion, Terza et al. (2008)
 - Ordinal models: Two-Stage Least Squares, Angrist (1998)



Spatial filters by mode.

Modelling results: Linear regressions

Calculated accessibility	Labour force		Employed population	
	Few jobs available	Declining employments	Too long to reach work	Easy to reach work
Active		13 (**)		
- For active travellers	-12 (*)			
Public transit	-76 (***)			
- For transit users			-213 (*)	
Motorized				
R-squared	0.24	0.40	0.21	0.09

Accessibility coefficients for calculated accessibility indicators in the linear models
(Signif. Codes (p-value): 0 '***' 0.001 '**' 0.01 '*' 0.05).

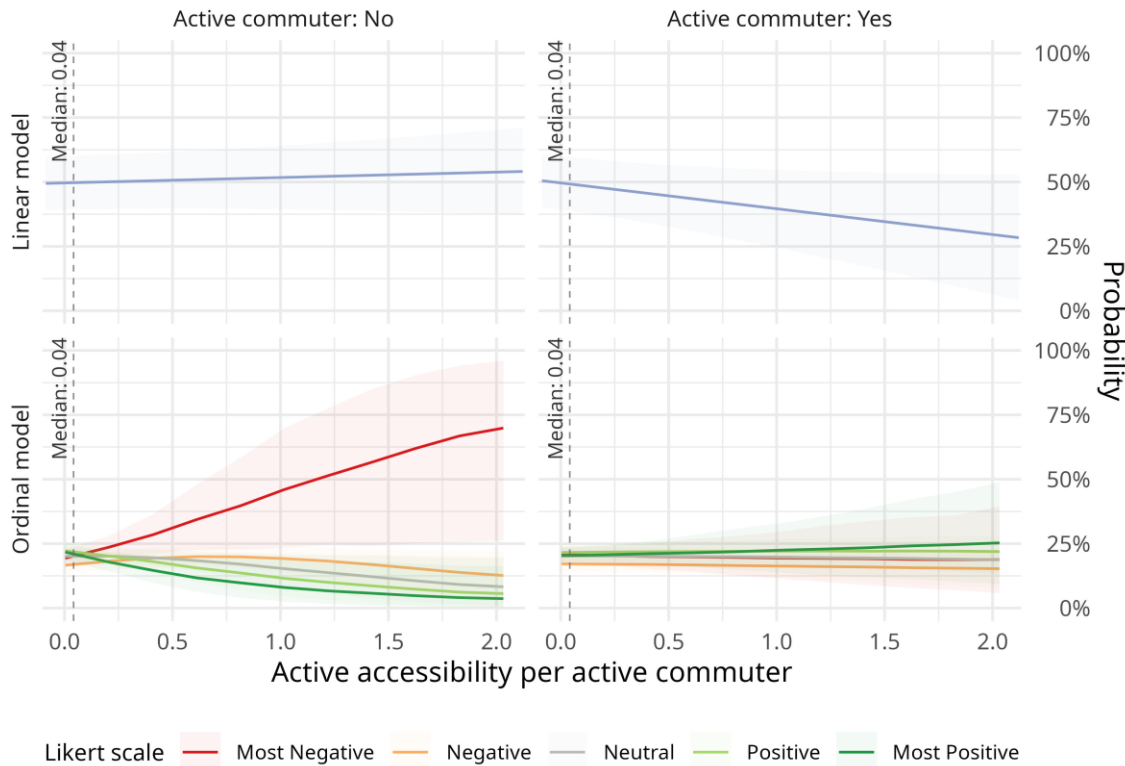
Modelling results: Ordinal regressions

Calculated accessibility	Labour force		Employed population	
	Few jobs available	Declining employments	Too long to reach work	Easy to reach work
Active	-14% (*)	-16% (*)		
- For active travellers	17% (*)	21% (*)		
Public transit	25% (*)			
- For transit users				
Motorized				

Accessibility Odds ($\text{Exp}(\beta)$) for calculated accessibility indicators in the ordinal categorical models
 (Signif. Codes (p-value): 0 '***' 0.001 '**' 0.01 '*' 0.05).

Modelling results: Active accessibility

"There are too few jobs available for me in my ideal commuting time"



Effect of calculated job accessibility on perceived employment reachability (p-value < 0.05).

Calculated

Active

- for active travellers

Linear outcome

Non-significative

Satisfaction (-12)*

Calculated

Active

- active travellers

Ordinal outcome

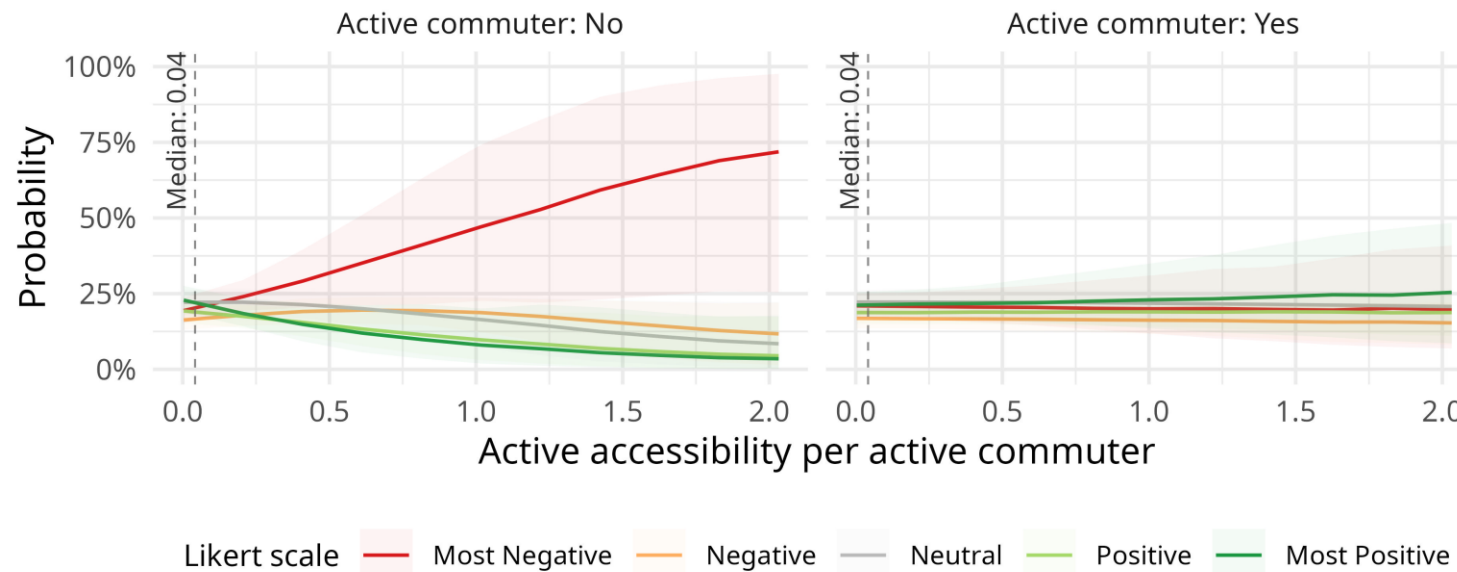
Dissatisfaction (-14%)* **Net:**

Satisfaction (+17%)* **+0.62%**

Significant codes: 0 '****' 0.001 '***' 0.01 '**' 0.05

Modelling results: Active accessibility

"I have had to decline employment opportunities because of my transport situation"

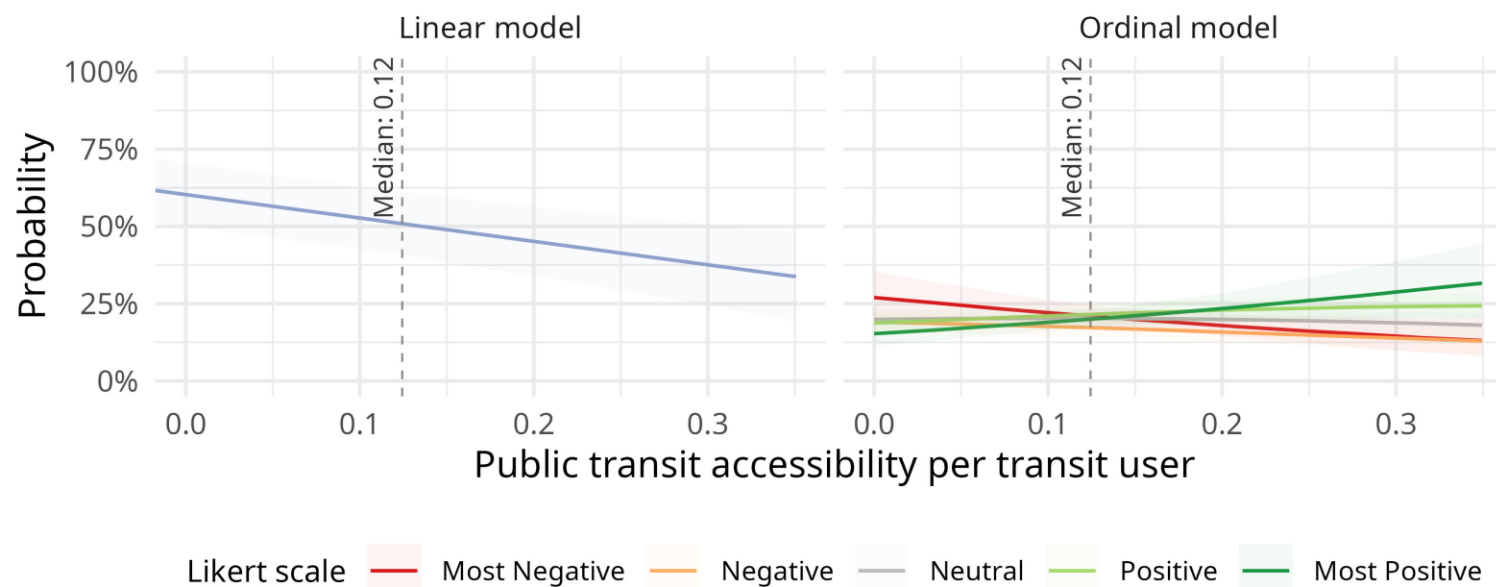


Calculated	Linear model	Ordinal model
Active	Dissatisfaction (-13)**	Dissatisfaction (-16%)*
- for active commuters	Non-significative	Satisfaction (+21%)*
		Net: +1.64%

Effect of calculated job accessibility on perceived employment reachability (p-value < 0.05).

Modelling results: Transit accessibility

"There are too few jobs available for me in my ideal commuting time"



Calculated

Linear model

Ordinal model

Public Transit

Satisfaction (-76)***

Satisfaction (+25%)*

- for transit users

Non-significative

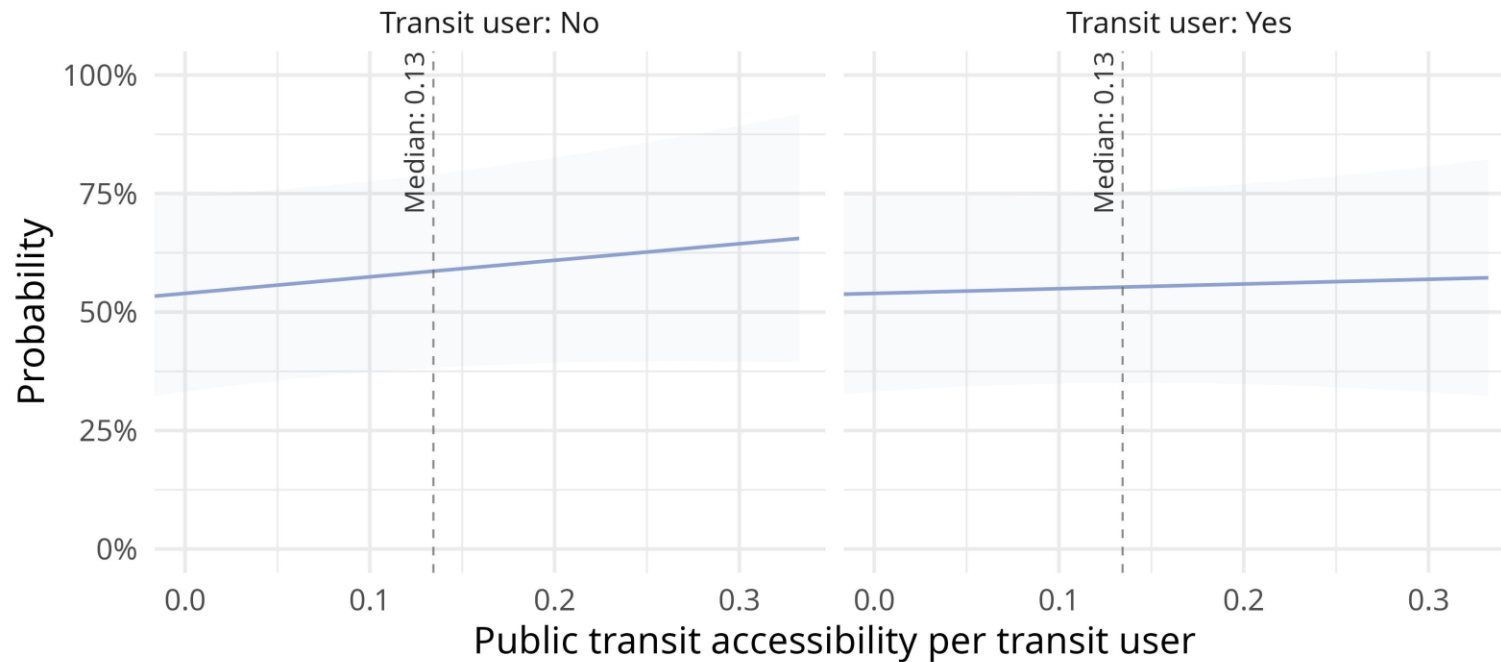
Non-significative

Effect of calculated job accessibility on perceived employment reachability (p-value < 0.05).

Significant codes: 0 '****' 0.001 '***' 0.01 '**' 0.05

Modelling results: Transit accessibility

"It takes too long to travel to work"



Calculated

Public Transit

- *for transit users*

Linear model

Dissatisfaction (231)

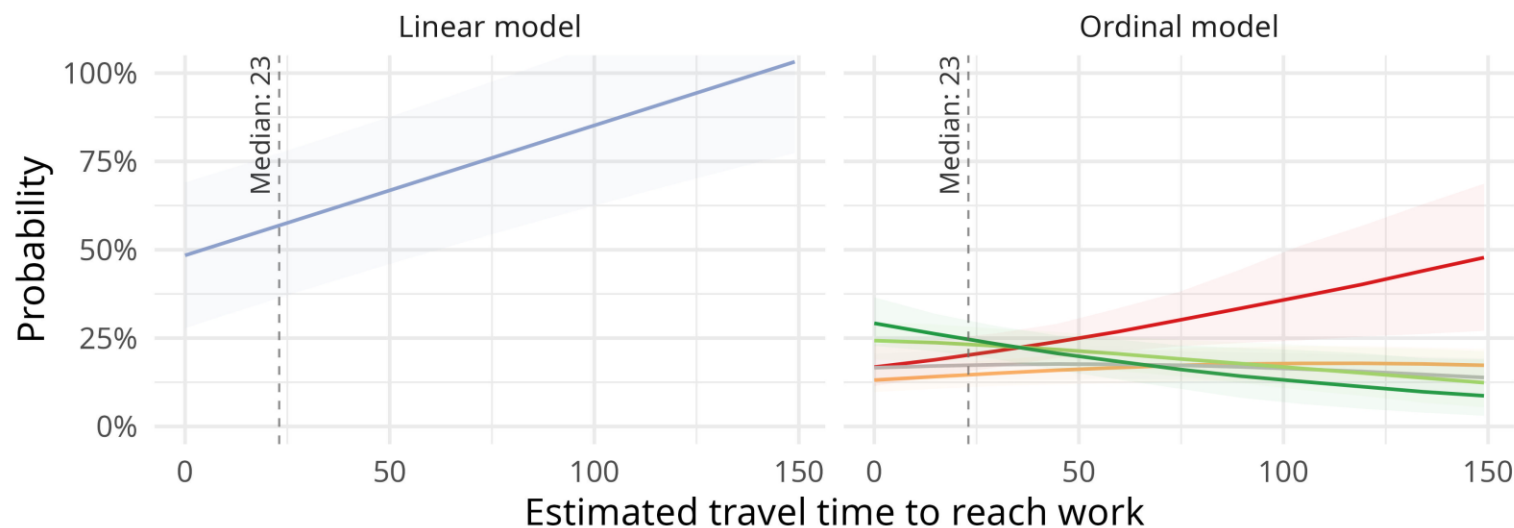
Satisfaction (-213)*

Effect of calculated job accessibility on perceived employment reachability (p-value < 0.05).

Significant codes: 0 '****' 0.001 '***' 0.01 '**' 0.05

Modelling results: Commuting time

"It takes too long to travel to work"



Likert scale — Most Negative — Negative — Neutral — Positive — Most Positive

Calculated	Linear model	Ordinal model
Travel time to reach work	Dissatisfaction (-0.37)***	Dissatisfaction (-4%)*

Effect of calculated job accessibility on perceived employment reachability (p-value < 0.05).

Significant codes: 0 '****' 0.001 '***' 0.01 '**' 0.05

Conclusion

- Motorized accessibility was not statistically significant for any outcomes
- Active accessibility was associated with a reduction in satisfaction for two outcomes, but this negative impact was generally offset for active commuters
- Transit accessibility has a great effect on the perception of having too few jobs available in an ideal commuting time, as detected in both types of models
- Calculated accessibility is significant for models based on the entire labour force

Thanks!

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Github: github.com/dias-bruno/NSTP/



**MOBILIZING
JUSTICE** —

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